**MIDTERM EXAMINATION**
**Spring 2010**
**MTH301- Calculus II (Session - 2)**

**Ref No: 1499740**
**Time: 60 min**
**Marks: 40**

| **Student Info** | |
|---|---|
| **StudentID:** | **BC080402259** |
| **Center:** | **OPKST** |
| **ExamDate:** | **6/4/2010 12:00:00 AM** |

| **For Teacher's Use Only** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Q No.** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **Total** |
| **Marks** | | | | | | | | | |
| **Q No.** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | |
| **Marks** | | | | | | | | | |
| **Q No.** | **17** | **18** | **19** | **20** | **21** | **22** | **23** | **24** | |
| **Marks** | | | | | | | | | |
| **Q No.** | **25** | **26** | | | | | | | |
| **Marks** | | | | | | | | | |

**Question No: 1 ( Marks: 1 ) - Please choose one**

Which of the following is the interval notation of real line?

- ► $(-\infty , +\infty)$
- ► $(-\infty , 0)$
- ► $(0 , +\infty)$

**Question No: 2 ( Marks: 1 ) - Please choose one**

What is the general equation of parabola whose axis of symmetry is parallel to y-axis?

- ► $y = ax^2 + b \quad (a \neq 0)$
- ► $x = ay^2 + b \quad (a \neq 0)$
- ► $y = ax^2 + bx + c \quad (a \neq 0)$
- ► $x = ay^2 + by + c \quad (a \neq 0)$

**Question No: 3 ( Marks: 1 ) - Please choose one**

Which of the following is geometrical representation of the equation $y = 4$ , in three dimensional space?

- ► A point on y-axis
- ► Plane parallel to xy-plane
- ► Plane parallel to yz-axis
- ► Plane parallel to xz-plane

**Question No: 4 ( Marks: 1 ) - Please choose one**

Suppose $f(x, y) = x^3 e^{xy}$ . Which one of the statements is correct?

- ► $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$
- ► $\frac{\partial f}{\partial y} = x^3 e^{xy}$
- ► $\frac{\partial f}{\partial y} = x^4 e^{xy}$

- ▶ $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

**Question No: 5 ( Marks: 1 ) - Please choose one**

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

- ▶ $2xy + \frac{1}{x^2}$
- ▶ $2y + \frac{1}{x^2}$
- ▶ $2xy - \frac{1}{x^2}$
- ▶ $2y - \frac{1}{x^2}$

**Question No: 6 ( Marks: 1 ) - Please choose one**

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$\frac{\partial w}{\partial r} =$

- ▶ $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$
- ▶ $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- ▶ $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- ▶ $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

**Question No: 7 ( Marks: 1 ) - Please choose one**

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} \frac{3x^2 y}{x^2 + y^2} & if\ (x, y) \neq 0 \\ 0 & if\ (x, y) = 0 \end{cases}$$

- ▶ $f(x, y)$ is continuous at origin

- ► $f(0, 0)$ is not defined
- ► $f(0, 0)$ is defined but $\lim_{(x,y)\to(0, 0)} f(x, y)$ does not exist
- ► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0, 0)} f(x, y)$ exists but these two numbers are not equal.

**Question No: 8 ( Marks: 1 ) - Please choose one**

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

- ► Area of R.
- ► Radius of inscribed circle in R.
- ► Distance between two endpoints of R.
- ► None of these

**Question No: 9 ( Marks: 1 ) - Please choose one**

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are ---------

- ► Parallel
- ► Perpendicular
- ► In opposite direction

**Question No: 10 ( Marks: 1 ) - Please choose one**

Two surfaces are said to intersect orthogonally if their normals at every point common to them are ----------

- ► perpendicular
- ► parallel
- ► in opposite direction

**Question No: 11 ( Marks: 1 ) - Please choose one**

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point $(x_0, y_0)$ and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) > 0$ then $f$ has ----------

- ► Relative maximum at $(x_0, y_0)$
- ► Relative minimum at $(x_0, y_0)$
- ► Saddle point at $(x_0, y_0)$

► No conclusion can be drawn.

**Question No: 12 ( Marks: 1 ) - Please choose one**

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Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point $(x_0, y_0)$ and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then $f$ has ----------

► Relative maximum at $(x_0, y_0)$

► Relative minimum at $(x_0, y_0)$

► Saddle point at $(x_0, y_0)$

► No conclusion can be drawn.

**Question No: 13 ( Marks: 1 ) - Please choose one**

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Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point $(x_0, y_0)$ and let $D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D < 0$ then $f$ has ----------

► Relative maximum at $(x_0, y_0)$

► Relative minimum at $(x_0, y_0)$

► Saddle point at $(x_0, y_0)$

► No conclusion can be drawn

**Question No: 14 ( Marks: 1 ) - Please choose one**

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Let $(x_1, y_1, z_1)$ and $(x_2, y_2, z_2)$ be any two points in three dimensional space. What does the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ calculates?

► Distance between these two points

► Midpoint of the line joining these two points

► Ratio between these two points

**Question No: 15 ( Marks: 1 ) - Please choose one**

The function $f(x, y) = \sqrt{y - x}$ is continuous in the region --------- and discontinuous elsewhere.

- $x \neq y$
- $x \leq y$
- $x > y$

**Question No: 16 ( Marks: 1 ) - Please choose one**

Plane is an example of ----------

- Curve
- Surface
- Sphere
- Cone

**Question No: 17 ( Marks: 1 ) - Please choose one**

If $R = R_1 \cup R_2$ , where $R_1$ and $R_2$ are no overlapping regions then

$$\iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA =$$

- $\iint_{R} f(x,y)dA$
- $\iint_{R_1} f(x,y)dA \cup \iint_{R_2} f(x,y)dA$
- $\iint_{R} f(x,y)dV$
- $\iint_{R_1} f(x,y)dA \cap \iint_{R_2} f(x,y)dA$

**Question No: 18 ( Marks: 1 ) - Please choose one**

$If\ R = \{(x,y) / 0 \leq x \leq 2\ and\ 1 \leq y \leq 4\},\ then$

$$\iint_{R} (6x^2 + 4xy^3)dA =$$

- $\int_{1}^{4}\int_{0}^{2}(6x^2+4xy^3)dydx$
- $\int_{0}^{2}\int_{1}^{4}(6x^2+4xy^3)dxdy$
- $\int_{1}^{4}\int_{0}^{2}(6x^2+4xy^3)dxdy$
- $\int_{2}^{4}\int_{0}^{1}(6x^2+4xy^3)dxdy$

**Question No: 19 ( Marks: 1 ) - Please choose one**

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$If\ R=\{(x,y)/0\le x\le 2\ and\ 0\le y\le 3\},then$
$\iint_R (1-ye^{xy})dA=$

- $\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dydx$
- $\int_{0}^{2}\int_{0}^{3}(1-ye^{xy})dxdy$
- $\int_{2}^{3}\int_{0}^{0}(1-ye^{xy})dxdy$
- $\int_{0}^{2}\int_{2}^{3}(4xe^{2y})dydx$

**Question No: 20 ( Marks: 1 ) - Please choose one**

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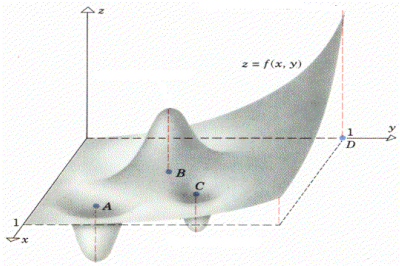
$If\ R=\{(x,y)/0\le x\le 4\ and\ 0\le y\le 9\},then$
$\iint_R (3x-4x\sqrt{xy})dA=$

- $\int_{0}^{9}\int_{0}^{4}(3x-4x\sqrt{xy})dydx$
- $\int_{0}^{4}\int_{4}^{9}(3x-4x\sqrt{xy})dxdy$
- $\int_{4}^{9}\int_{0}^{0}(3x-4x\sqrt{xy})dxdy$

▶ $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy})dydx$

**Question No: 21 ( Marks: 2 )**

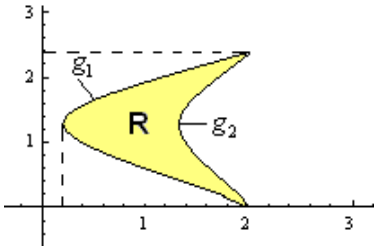
Following is the graph of a function of two variables




In its whole domain, state whether the function has relative maximum value or absolute maximum value at point B. Also, justify your answer

**Question No: 22 ( Marks: 2 )**

Let the function $f(x, y)$ is continuous in the region R, where R is bounded by graph of functions $g_1$ and $g_2$ (as shown below).




In the following equation, replace question mark (?) with the correct value.

$$\iint_R f(x, y)\, dA = \int_?^? \int_?^? f(x, y)\ _____?____$$

**Question No: 23 ( Marks: 3 )**

Evaluate the following double integral.

$\iint (3 + 2x - 3y^2)\, dx\, dy$

**Question No: 24 ( Marks: 3 )**

*Let* $f(x, y, z) = yz^3 - 2x^2$

*Find the gradient of* $f$.

**Question No: 25 ( Marks: 5 )**

Find, Equation of Normal line (in parametric form) to the surface $f(x, y, z) = xy + 2yz - xz^2 + 10$ at the point $(-5, 5, 1)$

**Question No: 26 ( Marks: 5 )**

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x$ and $y = x^2$, in the first quadrant.

